PLUGGING INTO THE FUTURE

An Exploration of Electricity Consumption Patterns using Machine Learning

Document:	Project Documentation	**Date:**	July 14, 2026
Domain:	Data Science & Web Development	**Technology:**	Python (Flask), ML, SQLite

1. Project Description

The **Electricity Consumption Patterns Exploration** is a Machine Learning-based web application developed to predict whether a consumer's daily electricity usage is likely to be High, Medium, or Low based on weather, time, and historical consumption information. The application assists utility providers and energy grid managers in making quick and accurate load management decisions while reducing energy wastage.

The system is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, SQLite, HTML, and CSS**. It utilizes a trained Random Forest Classifier to analyze consumption details such as peak hours, temperature, humidity, household size, appliance types, and other environmental attributes.

The application includes a secure user authentication system where authorized users can log in before accessing the prediction system. After successful authentication, users can enter consumption details through an easy-to-use web interface. The machine learning model processes the information and instantly predicts whether the consumption is a High Usage (Alert) or Normal Usage (Efficient) along with a confidence score.

Every prediction is automatically stored in a SQLite database, allowing users to view previous usage records through the Prediction History page. The application also provides a Dashboard that displays consumption analytics such as total load processed, peak demand hours, average consumption rate, prediction accuracy, and overall model statistics.

2. Project Scenarios

Scenario 1: Electricity Consumption Prediction

An authorized user logs into the system and enters details such as weather conditions, peak hours, and household size. The trained Random Forest model analyzes the information and predicts whether the electricity consumption is likely to be High or Normal, along with the prediction confidence score.

Scenario 2: Prediction History Management

After every prediction, the consumption details, prediction result, confidence score, date, and time are automatically stored in the SQLite database. Users can access the Prediction History page to review previous prediction records whenever required.

Scenario 3: Dashboard Analytics

The Dashboard retrieves prediction data from the database and displays important analytics such as total predictions, high consumption alerts, normal consumption records, high usage rate, average prediction confidence, and total records processed. This enables users to monitor the overall performance of the prediction system.

Scenario 4: Secure User Authentication

Only registered users with valid login credentials can access the Electricity Consumption Prediction System. The authentication module verifies user credentials before allowing access to the prediction system, history records, and dashboard functionalities.

3. Technical Architecture (Database Schema)

The relational schema of the database used to handle application data, user profiles, and logs is detailed below:

Table Name	Core Columns	Description
USERS	UserID (PK), Username, Password, Email, CreatedAt	Stores registered web application users and credentials.
CONSUMPTION_DETAILS	RecordID (PK), UserID (FK), Region, HouseholdSize, Temperature, Humidity, PeakHours, DayOfWeek	Stores the input parameters submitted for model prediction.
USAGE_PREDICTION	PredictionID (PK), RecordID (FK), ModelID (FK), PredictionResult, ConfidenceScore, PredictionDate	Stores output predictions and confidence score metrics.
ML_MODEL	ModelID (PK), ModelName, Algorithm, Accuracy, TrainingDate, ModelFilePath	Maintains metadata about current and past active ML models.

4. Performance Testing Report

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter
Type of Testing	Load Testing and Stress Testing
Target Module/API	Electricity Consumption Prediction Model & Dashboard API
Test Environment	Local Host (Development Server)

Step 2: Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (s)	Expected Outcome
1	**User Login:** Concurrent session logins.	50	60	Secure login under 1.5s.
2	**Consumption Prediction:** Send weather/ peak-load variables.	100	120	ML output returns under 2s.
3	**Dashboard Load:** Render charts and visual summaries.	75	180	Graphs display under 2.5s.

Step 3: Performance Test Results

Metric	Target Value	Actual Value	Status
Response Time (Avg)	< 2 seconds	1.12 seconds	**Pass**
Response Time (Max)	< 5 seconds	3.45 seconds	**Pass**
Throughput	> 50 req/sec	68.4 req/sec	**Pass**
Error Rate	< 1%	0.00%	**Pass**

5. File / Folder Structure

```
electricity-consumption-analysis/
├── app.py                       # Main Flask application file
├── database.db                  # SQLite Database storing predictions and user
history
├── requirements.txt             # Python dependencies (Flask, scikit-learn, etc.)
├── README.md                    # Setup guide and instructions to run the project
├── model/
│   └── power_consumption_model.pkl  # Trained Random Forest ML model file
├── static/
│   ├── css/
│   │   └── style.css            # Custom CSS for styling the dashboard
│   └── js/
│       └── dashboard_charts.js # JavaScript code to render consumption graphs
└── templates/
    ├── login.html               # Authentication page
    ├── index.html               # Main prediction form page
    ├── dashboard.html           # Analytics & charts rendering dashboard
    └── history.html             # Past prediction logs and SQLite history view page
```

6. Submission Checklist & Access Details

S.No	Item to Submit	Status
1	Complete source code (all files and folders)	**Yes**
2	README / Setup Guide	**Yes**
3	dependencies (requirements.txt)	**Yes**
4	Database schema / SQLite files	**Yes**
5	Test files and performance reports	**Yes**

Access Credentials:

- **Local URL:** `http://localhost:5000`
- **Demo User:** `admin`

- **Demo Pass:** `admin123`